

Agent-Based Modeling and Equation-Based Modeling of Rift Valley Fever Virus Transmission: An Ecosystem-Integrated Scientific Literature Review

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Abstract

Rift Valley fever (RVF) is a vector-borne zoonotic disease that poses significant threats to livestock and human health across Africa and the Arabian Peninsula. This comprehensive literature review synthesizes scientific evidence on agent-based modeling (ABM) and equation-based modeling (EBM) approaches to understanding RVF virus (RVFV) transmission, with particular emphasis on ecosystem-integrated studies. Drawing from 372 unique papers identified across SciSpace, Google Scholar, PubMed, and ArXiv, this review examines methodological frameworks, mathematical analyses, and ecosystem integration strategies.

ABM approaches have utilized platforms such as CORMAS and custom multi-agent systems to capture spatial heterogeneity, herd mobility, and vector-host interactions, particularly in the Ferlo region of Senegal. EBM approaches have employed compartmental ordinary differential equations (ODEs), network meta-population models, delay differential equations (DDEs), and fractional-order formulations to analyze transmission dynamics, compute basic reproduction numbers (R_0), and evaluate intervention strategies. Ecosystem-integrated studies have combined field entomology, hydrological modeling, and high-resolution remote sensing (TRMM, TerraSAR-X, SPOT-5) to link pond dynamics, rainfall patterns, and vector ecology with transmission risk.

Key findings indicate that vector biting rates, mosquito mortality, herd mobility, and vertical transmission are primary drivers of RVF outbreaks. Vaccination strategies, vector control, and movement restrictions show promise as interventions. Significant research gaps remain in integrating ABM and EBM approaches, incorporating climate change projections, and developing unified frameworks that capture both local heterogeneity and regional spread dynamics.

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1. Introduction

Rift Valley fever (RVF) is a vector-borne viral hemorrhagic disease caused by the RVF phlebovirus (RVFV), classified within the family *Phenuiviridae*. It primarily affects domestic ruminants—cattle, sheep, goats, and camels—but can cause severe illness in humans, ranging from mild febrile illness to hemorrhagic fever, encephalitis, and retinitis, with case-fatality rates reaching up to 50% in severe human cases [Cécilia et al., 2022]. First described in the Rift Valley of Kenya in 1931, RVF has since been documented across sub-Saharan Africa, North Africa, and the Arabian Peninsula, with ongoing expansion risk to Europe and North America through competent vector species [Xue et al., 2013].

The disease’s epidemic nature is driven by complex interactions between climatic conditions, vector ecology, and livestock management practices. Explosive outbreaks typically follow periods of above-normal rainfall that create extensive breeding habitats for floodwater mosquitoes, particularly *Aedes vexans* and *Culex poicilipes* in West Africa [Durand et al., 2020]. Between epidemics, the virus circulates at low levels, maintained through vertical transmission in *Aedes* mosquitoes and continuous low-level transmission among livestock [Pedro et al., 2016].

Computational modeling has become an indispensable tool for understanding RVFV transmission dynamics and evaluating control strategies. Two principal modeling paradigms have emerged:

- **Agent-Based Modeling (ABM):** Bottom-up approaches that represent individual entities (animals, mosquitoes, humans) as autonomous agents with specific attributes and behaviors, capturing spatial heterogeneity and stochastic processes.
- **Equation-Based Modeling (EBM):** Top-down mathematical frameworks using differential equations (ODEs, DDEs) to represent population-level dynam-

ics, enabling analytical computation of epidemic thresholds (R_0) and stability properties.

This review synthesizes the current state of knowledge on both ABM and EBM approaches for RVFV transmission, with particular attention to studies that have integrated ecosystem data from the Ferlo pastoral ecosystem of northern Senegal, the Sahelian belt, and East African landscapes. We examine methodological innovations, key findings on transmission dynamics, the role of environmental factors, and identify critical gaps that should guide future research.

2. Background and Theoretical Foundations

2.1. RVFV Epidemiology and Transmission Cycle

RVFV is maintained in nature through a complex multi-host, multi-vector transmission cycle. In West Africa, the two principal mosquito vectors are *Aedes vexans arabiensis* (floodwater mosquito) and *Culex poicilipes* (secondary amplification vector). These species have complementary ecological niches: *Aedes vexans* emerges explosively following rainfall events that flood temporary ponds, while *Culex poicilipes* breeds in more permanent water bodies and maintains populations throughout the year [Soti et al., 2012].

A critical epidemiological feature of RVFV is transovarial (vertical) transmission in *Aedes* mosquitoes, whereby infected females deposit virus-bearing eggs that can survive desiccation for years in soil, hatching when flooded to produce infected adults. This mechanism is hypothesized to explain inter-epidemic persistence, though its relative epidemiological importance versus continuous reintroduction by nomadic herds remains debated [Durand et al., 2020].

2.2. The Ferlo Ecosystem as a Model System

The Ferlo region of northern Senegal has emerged as a focal point for RVFV ecosystem research. This semi-arid Sahelian ecosystem, characterized by annual rainfall of 300–500 mm concentrated in a July–October rainy season, supports a dense network of temporary ponds (*mares*) that drive both vector ecology and livestock distribution [Paul et al., 2014]. The coexistence of sedentary resident herds and nomadic transhumant herds creates a natural experiment for studying how animal movements shape transmission dynamics.

2.3. Rationale for Computational Modeling

The complex, multi-scale nature of RVFV transmission—spanning individual mosquito biting events, herd-level immunity, pond-scale vector dynamics, and regional climate patterns—makes computational modeling essential. ABM and EBM approaches offer complementary strengths: ABM captures individual heterogeneity and spatial structure, while EBM enables analytical tractability and population-level parameter estimation [Cécilia et al., 2022].

3. Agent-Based Modeling Methodologies

3.1. CORMAS Platform and Multi-Agent Systems

The most extensively documented ABM framework for RVFV in the Ferlo ecosystem uses the CORMAS (Common-pool Resources and Multi-Agent Systems) platform. Developed by CIRAD, CORMAS provides a Smalltalk-based environment for spatial multi-agent simulation with explicit representation of landscape, resources, and agent behaviors.

Paul et al. [2014] developed the foundational Ferlo ABM using CORMAS, representing the following agent classes:

- **Herd agents:** Cattle and small ruminant herds with location, immunity status, and mobility behavior attributes
- **Pond agents:** Temporary water bodies with rainfall-driven volume dynamics and associated mosquito populations
- **Village/camp agents:** Spatial nodes representing sedentary settlements and nomadic camp locations
- **Vector agents:** Mosquito populations associated with ponds, with infection status and biting behavior

The spatial grid used approximately 100 m² cell resolution with daily time steps, typical simulation periods of 30 days, and parameter sweeps over rainfall scenarios, herd density, and mobility patterns. This framework was subsequently extended to examine rainfall impacts on transmission [Paul et al., 2018], demonstrating that strong rains amplify transmission by increasing pond volumes and mosquito multiplication.

3.2. Individual-Based Network Models

Beyond the Ferlo-specific CORMAS framework, individual-based network models have been applied to RVFV at larger spatial scales. Niu et al. [2012] developed a spatially explicit epidemiological model incorporating livestock movement networks across subcounties, with vector abundance parameterized by environmental suitability layers. This approach enables evaluation of movement-based interventions such as livestock quarantine and trade restrictions.

A key finding from network-based ABM is the “seeding paradox”: fewer initial infectious animals can paradoxically produce larger, more geographically dispersed outbreaks because low initial seeding allows the virus to spread widely before amplification, encountering naïve populations across multiple nodes [Xue et al., 2013].

3.3. UML Conceptual Modeling Approaches

A methodological innovation in RVFV ABM is the use of Unified Modeling Language (UML) to formalize conceptual models prior to computational implementation.

Paul et al. [2018] demonstrated that UML static diagrams improve model design traceability, facilitate scenario definition, and enable systematic documentation of agent attributes, relationships, and behavioral rules. This practice enhances model reproducibility and facilitates comparison across studies.

3.4. Spatial Simulation and Herd Mobility

Herd mobility is a primary focus of RVFV ABM, reflecting the critical role of transhumance in Sahelian pastoral systems. Key findings from spatial ABM include:

- **Mobility amplification:** Paul et al. [2014] demonstrated up to 33% variation in infection rates between no-mobility and high-mobility scenarios in the Ferlo, with mobility increasing exposure at high-density pond aggregation points.
- **Rainfall–mobility interaction:** Delayed rains and dry spells interact with herd movement patterns to alter outbreak timing and magnitude; sensitivity analyses show these interactions are non-linear [Paul et al., 2018].
- **Nomadic introduction:** Durand et al. [2020] showed that yearly introductions by nomadic herds alone can explain endemic RVFV circulation in northern Senegal, without requiring vertical transmission.
- **Control implications:** Restricting cattle movements during peak vector abundance periods reduces spread in simulation, supporting movement-based interventions as a complement to vaccination [Gachohi et al., 2016].

4. Equation-Based Modeling Methodologies

4.1. Compartmental SIR/SEIR Models

Compartmental ODE models form the backbone of EBM for RVFV, typically incorporating susceptible (S), exposed (E), infectious (I), and recovered (R) compartments for both vertebrate hosts and mosquito vectors. The general multi-host, multi-vector framework takes the form:

$$\frac{dS_H}{dt} = \Lambda_H - \beta_{VH} \frac{I_V}{N_V} S_H - \mu_H S_H \quad (1)$$

$$\frac{dI_H}{dt} = \beta_{VH} \frac{I_V}{N_V} S_H - (\gamma_H + \mu_H) I_H \quad (2)$$

$$\frac{dS_V}{dt} = \Lambda_V - \beta_{HV} \frac{I_H}{N_H} S_V - \mu_V S_V \quad (3)$$

$$\frac{dI_V}{dt} = \beta_{HV} \frac{I_H}{N_H} S_V - \mu_V I_V \quad (4)$$

where β_{VH} and β_{HV} are vector-to-host and host-to-vector transmission rates, Λ denotes recruitment rates, μ denotes mortality rates, and γ_H is the host recovery rate. Extensions include multiple host species (cattle, sheep, goats, humans), multiple vector species, exposed classes, and vertical transmission terms [Mpeshe et al., 2011, Cécilia et al., 2022].

Mpeshe et al. [2011] developed a comprehensive mathematical model of RVFV with explicit human host compartments, deriving R_0 via the next-generation matrix method and performing sensitivity analysis to identify key transmission parameters. Their analysis confirmed that mosquito biting rate and mortality are the dominant determinants of epidemic potential.

4.2. Network and Meta-Population Models

Recognizing that RVFV transmission occurs across multiple spatial scales, several studies have developed network and meta-population ODE frameworks. Xue et al. [2013] proposed a hierarchical network model representing transmission within local populations (nodes) as ODE systems and between populations through animal movement links (edges). This framework enables simulation of regional spread from local outbreaks and evaluation of movement-based control measures.

Gao et al. [2013] modeled the spatial spread of RVFV in Egypt using a patch model with livestock movement between governorates, demonstrating that spatial heterogeneity in vector abundance and host density produces complex spatial patterns

of epidemic spread not captured by homogeneous models.

4.3. Delay Differential Equations

Delay differential equation (DDE) formulations have been applied to capture temporal lags in RVFV transmission dynamics. These include:

- **Incubation delays:** Extrinsic incubation period (EIP) in mosquitoes before they become infectious
- **Intrinsic incubation:** Latency period in vertebrate hosts
- **Immune response delays:** Time to protective immunity following infection or vaccination

Raza et al. [2026] employed entropy-based analysis of RVFV DDE models, using LaSalle invariance principles to establish global stability conditions for the disease-free equilibrium. Their results showed that delay formulations can produce qualitatively different dynamics from ODE equivalents, including oscillatory behavior and delayed epidemic peaks.

4.4. Fractional-Order Models

Fractional-order differential equation (FODE) models have been explored as alternatives to integer-order ODEs for RVFV, motivated by their ability to capture memory effects and anomalous diffusion in disease dynamics. Oguntolu et al. [2022] developed a fractional-order RVFV model with human host, demonstrating that the fractional order parameter α modulates epidemic trajectories and that $\alpha = 1$ recovers the classical ODE limit. While mathematically elegant, FODE models face challenges in biological interpretation of the fractional order and in parameter estimation from field data.

5. Mathematical Analysis Methods

5.1. R_0 Computation and Threshold Analysis

The basic reproduction number R_0 —the expected number of secondary infections generated by a single infectious individual in a fully susceptible population—is the central analytical output of EBM for RVFV. For a two-host (cattle C , mosquito V) system, R_0 is typically derived via the next-generation matrix method as:

$$R_0 = \sqrt{\frac{a^2 b_C b_V N_V}{\mu_V (\gamma_C + \mu_C) N_C}} \quad (5)$$

where a is the mosquito biting rate, b_C and b_V are host-to-vector and vector-to-host transmission probabilities per bite, μ_V and μ_C are mosquito and host mortality rates, γ_C is the host recovery rate, and N_V/N_C is the vector-to-host ratio [Mpeshe et al., 2011].

This formulation reveals that R_0 scales with the square of the biting rate a , making biting rate reduction (via insecticides or repellents) disproportionately effective at reducing epidemic potential. The disease-free equilibrium is locally asymptotically stable when $R_0 < 1$ and unstable when $R_0 > 1$ [Cécilia et al., 2022].

5.2. Stability Analysis

Both local and global stability analyses have been performed for RVFV ODE models:

- **Local stability:** Assessed via linearization and Routh–Hurwitz criteria for the Jacobian matrix at equilibrium points. Results confirm the disease-free equilibrium (DFE) is locally stable when $R_0 < 1$ [Mpeshe et al., 2011].
- **Global stability:** Lyapunov function methods and LaSalle’s invariance principle have been applied to establish global stability of the DFE and endemic equilibrium under specific parameter conditions [Raza et al., 2026].
- **Delay systems:** For DDE formulations, characteristic equation analysis and entropy-based methods establish stability conditions involving both R_0 and delay magnitudes [Raza et al., 2026].

5.3. Sensitivity Analysis

Sensitivity and elasticity analyses identify the model parameters with greatest influence on R_0 and endemic prevalence, guiding intervention prioritization. Across RVFV EBM studies, consistent findings include:

1. **Mosquito biting rate** (a): Highest positive sensitivity in virtually all models, as $R_0 \propto a^2$ [Mpeshe et al., 2011, Cécilia et al., 2022]
2. **Mosquito mortality rate** (μ_V): Strong negative sensitivity; increasing adult mosquito mortality is the most efficient single-parameter intervention
3. **Vector-to-host ratio** (N_V/N_C): Positive sensitivity reflecting the importance of vector abundance relative to host density
4. **Vertical transmission rate**: Positive effect on R_0 and endemic prevalence, enabling persistence at lower vector densities [Pedro et al., 2016]

5.4. Optimal Control Formulations

Optimal control theory has been applied to RVFV models to identify cost-minimizing intervention strategies. Formulations typically minimize an objective functional:

$$J(u) = \int_0^T \left[w_1 I_H(t) + w_2 I_V(t) + \frac{1}{2} \sum_k c_k u_k^2(t) \right] dt \quad (6)$$

subject to the ODE system, where $u_k(t)$ are time-varying control variables (vaccination rate, insecticide application, livestock movement restrictions) and w_i , c_k are weighting coefficients [Gachohi et al., 2016]. Pontryagin’s maximum principle yields necessary conditions for optimal controls, solved numerically via forward-backward sweep methods. Results consistently recommend **pre-outbreak** vaccination campaigns over reactive post-detection vaccination, with optimal intensity depending on baseline R_0 and vaccine efficacy.

6. Ecosystem-Integrated Studies

6.1. Ferlo Ecosystem (Senegal)

The Ferlo region of northern Senegal (centered on Barkedji) has been the most intensively studied ecosystem for RVFV, offering a natural laboratory where endemic circulation, well-characterized ecology, and coexistence of resident and nomadic livestock systems enable detailed mechanistic study.

6.1.1. Pond Hydrology and Landscape Structure

Soti et al. [2010] demonstrated the feasibility of remote sensing and hydrological modeling for assessing spatio-temporal pond dynamics in the Ferlo. Their framework used:

- **TRMM rainfall data** as primary meteorological forcing for pond filling models
- **QuickBird imagery** for high-resolution pond delineation and boundary mapping
- **ASTER DEM** for catchment delineation and runoff routing
- **NDPI and NDTI indices** from SPOT-5 for pond area and water quality tracking

Strong agreement was achieved between modeled and measured pond water heights for larger ponds, establishing the tractability of hydrological modeling for RVF ecosystem studies.

Maura [2007] used SPOT-5 and QuickBird imagery to derive landscape indices (pond density, vegetation closure, perimeter-to-area ratios) that correlated with serological incidence in sheep, enabling identification of high-risk landscape configurations and spatial mapping of zones potentially occupied by mosquitoes (ZPOM).

6.1.2. Endemic Circulation and Serological Evidence

Durand et al. [2020] conducted a landmark study combining field serology (2015–2016) with a mechanistic epidemiological model at the pond level in the Younoufere area of northern Senegal. Key findings:

- IgG seroprevalence: 15.3% in resident herds ($n = 222$); 12.4% in nomadic herds ($n = 660$)
- IgM detection in 4% of nomadic herds during the second half of the rainy season, indicating recent transmission
- Nomadic herds arriving in the second half of the rainy season had predicted viraemic proportions of 3.4%
- Model comparison using ABC-APMC demonstrated nomadic introductions alone explain endemic circulation without requiring vertical transmission

6.1.3. Operational Risk Mapping

Vignolles et al. [2009] combined TRMM rainfall data, pond dynamics models, and measured vector aggressiveness (biting rates from human landing catches) to produce daily maps of exposure of parked ruminant herds—the first operational ZPOM-based risk mapping workflow for the Ferlo. These maps identified specific ponds and time windows of peak transmission risk, providing actionable information for pastoral communities and veterinary services.

6.2. Sahelian and East African Ecosystems

6.2.1. Sahelian Epidemic Potential Mapping

Soti et al. [2012] extended Ferlo-scale insights to the broader Sahel using a coupled hydrology–mosquito population model. Their framework mechanistically linked:

1. Daily TRMM rainfall $\rightarrow$ pond area and volume dynamics
2. Pond dynamics $\rightarrow$ larval habitat availability for *Aedes* and *Culex*

3. Larval habitat → adult mosquito abundance (via stage-structured population model)
4. Adult abundance → epidemic potential (R_0 proxy)

Results identified September as consistently the highest-risk month across Sahelian pixels, with spatial hotspots varying by year depending on rainfall distribution.

6.2.2. East African Ecosystems

East African RVFV outbreaks (Kenya, Tanzania, Somalia) differ from West African endemic patterns in their explosive, episodic nature following El Niño–Southern Oscillation (ENSO) events. [Barker et al. \[2013\]](#) developed a data-driven receptivity model for East Africa using environmental suitability layers (vegetation index, soil moisture, temperature) to map spatial variation in epidemic potential. [Eskew et al. \[2026\]](#) analyzed inter-epidemic transmission in East African hotspots, finding that hydrology remains the dominant risk driver and projecting that climate change will shift transmission risk geographically, with some currently endemic areas losing suitability while new areas become receptive.

6.3. Hydrology and Pond Dynamics

Hydrological modeling of temporary ponds is central to RVFV ecosystem studies across all geographic settings. The general modeling pipeline involves:

1. **Rainfall input:** TRMM or ground-station precipitation data at daily resolution
2. **Runoff routing:** DEM-based catchment delineation and runoff coefficients
3. **Pond filling/emptying:** Water balance equations accounting for infiltration and evapotranspiration
4. **Habitat suitability:** Pond area and persistence thresholds for larval development of target vector species

A critical insight from Ferlo hydrology studies is that the timing of dry spells within the rainy season—not just total rainfall—determines *Aedes vexans* emergence

pulses, because this species requires specific dry-wet cycles to trigger egg hatching (embryogenesis) [Soti et al., 2012]. This non-linear response to rainfall patterns means that simple rainfall thresholds are insufficient for early warning; temporal patterns of precipitation must be resolved.

6.4. Remote Sensing Technologies

Multiple remote sensing platforms have been integrated into RVFV ecosystem studies, each offering distinct capabilities:

Table 1: Remote sensing platforms used in RVFV ecosystem-integrated studies

Platform	Application	Key Studies
TRMM (Tropical Rainfall Measuring Mission)	Rainfall forcing for pond dynamics and vector population models	Soti et al. [2010, 2012], Vignolles et al. [2009]
SPOT-5 (optical, 5 m resolution)	Pond delineation, vegetation indices, landscape structure	Maura [2007], Soti et al. [2010]
QuickBird (optical, 0.6 m resolution)	High-resolution pond boundary mapping	Soti et al. [2010]
TerraSAR-X (X-band SAR)	All-weather free water detection; operational risk mapping	Vignolles et al. [2010]
MODIS	Regional vegetation and land surface temperature	Barker et al. [2013]
ASTER DEM	Catchment delineation for hydrological modeling	Soti et al. [2010]

The TerraSAR-X radar platform deserves particular mention as it enables weather-independent mapping of pond water surfaces at high resolution, overcoming the cloud-cover limitations of optical sensors during the rainy season when transmission risk is highest [Vignolles et al., 2010]. The operational warning system demonstrated by Vignolles et al. [2010] combined TerraSAR-X pond maps with TRMM rainfall and vector aggressiveness data to produce near-real-time risk bulletins for field use.

6.5. Vector Ecology: *Aedes vexans* and *Culex poicilipes*

The two principal RVFV vectors in West Africa have complementary ecological profiles that create distinct temporal patterns of transmission risk:

6.5.1. *Aedes vexans arabiensis*

- Floodwater mosquito; eggs laid in soil at pond margins during flooding
- Eggs require desiccation followed by flooding to hatch (obligate dry-wet cycle)
- Explosive emergence following suitable rainfall events; population peaks in early to mid-rainy season
- Capable of transovarial (vertical) RVFV transmission; eggs can remain viable for years in soil
- Host-feeding: Opportunistic, feeding on available livestock; documented feeding on cattle, sheep, goats, and humans in Ferlo [Biteye et al., 2019]

6.5.2. *Culex poicilipes*

- Breeds in more permanent water bodies; populations sustained throughout the rainy season
- Abundance positively correlated with cumulative rainfall and cooler minimum temperatures [Talla et al., 2014]
- Acts as a secondary amplification vector, maintaining transmission after initial *Aedes*-driven explosive phase
- No documented vertical transmission; maintenance depends on continued host-vector contact

This species complementarity creates a biphasic transmission pattern: an early explosive phase driven by *Aedes vexans* emergence and a sustained amplification phase maintained by *Culex poicilipes* throughout the rainy season [Soti et al., 2012]. Models that represent only one vector species underestimate transmission duration and may misidentify optimal intervention timing.

6.6. Climate Early Warning Systems

The integration of ecosystem data with transmission models has enabled development of operational early warning systems for RVF. The AdaptFVR project [Lafaye et al., 2013] exemplifies this approach, combining:

- Near-real-time TRMM rainfall monitoring
- Pond dynamics modeling with hydrological modules
- Entomological surveillance data for model calibration
- GPS-based herd location tracking for exposure assessment
- Risk bulletin generation for pastoral communities and veterinary services

The workflow produces daily ZPOM maps identifying ponds with active vector populations and herds potentially exposed, enabling targeted surveillance and preemptive interventions. Vignolles et al. [2010] demonstrated that TerraSAR-X imagery could be operationally integrated into this pipeline, providing weather-independent pond monitoring with 3–5 day revisit frequency.

7. Comparative Analysis: ABM vs. EBM Approaches

7.1. Methodological Strengths and Limitations

Table 2 summarizes the comparative strengths and limitations of ABM and EBM approaches for RVFV transmission modeling.

7.2. Convergent Findings Across Paradigms

Despite methodological differences, ABM and EBM studies converge on several key findings:

1. **Rainfall is the primary environmental driver:** Both ABM (Paul et al.) and EBM (Soti et al., Chemison et al.) identify rainfall as the dominant driver

Table 2: Comparative analysis of ABM and EBM approaches for RVFV transmission modeling

Criterion	Agent-Based Modeling (ABM)	Equation-Based Modeling (EBM)
Spatial representation	Explicit; individual locations tracked	Implicit (patch models) or absent
Individual heterogeneity	Full representation of individual variation	Population-averaged; limited heterogeneity
Stochasticity	Inherent; each run produces different outcomes	Deterministic (ODEs) or stochastic extensions
Analytical tractability	Low; requires simulation	High; analytical R_0 , stability, bifurcations
Computational cost	High; scales with population size	Low to moderate
Parameter estimation	Challenging; ABC methods required	Established MLE, Bayesian, OLS methods
Behavioral modeling	Natural; agent rules encode behavior	Difficult; requires structural extensions
Policy insights	Scenario-specific; high spatial resolution	General thresholds; R_0 -based interventions
Ecosystem integration	Natural; agents interact with landscape	Requires explicit coupling to environmental models
Validation	Difficult; pattern-oriented methods	Standard goodness-of-fit to incidence data

of RVFV outbreaks, operating through vector breeding habitat creation [Paul et al., 2014, Soti et al., 2012, Chemison et al., 2024].

- Vector parameters dominate R_0 :** EBM sensitivity analyses and ABM parameter sweeps consistently identify mosquito biting rate and mortality as the most influential parameters [Mpeshe et al., 2011, Paul et al., 2014].
- Herd mobility amplifies transmission:** ABM explicitly demonstrates mobility effects, while EBM network models show movement links accelerate regional spread [Paul et al., 2014, Xue et al., 2013].
- Seasonal amplification patterns:** Both paradigms identify late rainy season as the period of peak transmission, consistent with field serological data [Durand et al., 2020, Pedro et al., 2016].
- Vaccination is the most effective intervention:** Optimal control (EBM) and scenario analysis (ABM) both recommend pre-outbreak vaccination of young animals as the highest-impact single intervention [Gachohi et al., 2016,

Chemison et al., 2024].

7.3. Divergent Findings and Unresolved Questions

Several important questions show divergent findings across modeling paradigms:

1. **Role of vertical transmission:** EBM models that include vertical transmission show it can sustain endemic circulation; ABM studies (Durand et al.) show nomadic introductions alone suffice—the relative importance remains unresolved.
2. **Optimal spatial scale:** ABM favors fine-scale (pond-level) simulation; EBM network models operate at subcounty to national scales. The appropriate scale for operational decision-making is context-dependent.
3. **Wildlife reservoirs:** ABM studies have not systematically incorporated wildlife hosts; EBM models suggest buffalo and other ruminants may play roles in East African settings [Beechler, 2013].

8. Key Findings on Transmission Drivers

The synthesis of ABM, EBM, and ecosystem studies reveals a hierarchy of RVFV transmission drivers, ordered by strength of evidence and consistency across modeling paradigms:

1. **Vector biting rate and abundance** (highest evidence): Consistently the dominant parameter in R_0 sensitivity analyses; $R_0 \propto a^2$ implies biting rate reduction has quadratic impact on epidemic potential [Mpeshe et al., 2011].
2. **Rainfall and pond dynamics** (high evidence): Primary environmental driver linking climate to vector breeding habitat and abundance; temporal patterns of precipitation (not just totals) determine *Aedes* emergence pulses [Soti et al., 2012, Paul et al., 2014].

3. **Herd mobility and contact patterns** (high evidence): Transhumance and seasonal movements concentrate susceptible animals at high-vector-density water points; ABM quantifies 33% infection rate variation by mobility level [Paul et al., 2014].
4. **Host species composition** (moderate evidence): Cattle immunity reduces transmission more than equivalent immunity in small ruminants; multi-host models show species composition affects spatial risk patterns [Gachohi et al., 2016].
5. **Vertical transmission in *Aedes*** (moderate evidence): Enables low-level persistence and can amplify outbreaks when vector densities rise, but empirical efficiency estimates remain uncertain [Bron et al., 2023].
6. **Landscape structure** (moderate evidence): Pond density, vegetation closure, and connectivity shape vector-host contact rates; ZPOM mapping identifies high-risk landscape configurations [Vignolles et al., 2009].

9. Research Gaps and Future Directions

9.1. ABM–EBM Integration

A critical gap is the lack of frameworks that combine the spatial-behavioral richness of ABM with the analytical tractability of EBM. Hybrid approaches—such as ABM for local transmission coupled to EBM for regional spread—could leverage the strengths of both paradigms. The coupling framework proposed by Paul et al. [2020] represents a step in this direction, but systematic development and validation of hybrid models is needed.

9.2. Climate Change Projections

Current models have been validated against historical climate data, but projections of RVFV risk under climate change scenarios remain limited. Key needs include:

- Coupling of RVFV transmission models with General Circulation Model (GCM) outputs
- Assessment of how shifts in rainfall patterns and temperature affect vector phenology and transmission seasonality
- Identification of new geographic areas that may become receptive under future climate scenarios [Eskew et al., 2026]

9.3. Multi-Scale Dynamics

RVFV transmission occurs across scales from individual pond-level events to continental epidemic waves. Developing multi-scale modeling frameworks that link these levels—with ABM at local scales and EBM network models at regional scales—would improve both mechanistic understanding and operational forecasting capability [Xue et al., 2013].

9.4. Validation and Data Needs

Critical data gaps limiting model development and validation include:

1. **GPS-tracked animal movement data:** Systematic tracking of nomadic herd routes to replace assumptions in ABM mobility models
2. **Longitudinal vector abundance time series:** Multi-year, multi-site mosquito abundance data for model parameterization and validation
3. **Vertical transmission efficiency:** Experimental estimates of transovarial transmission rates under field-relevant conditions [Bron et al., 2023]
4. **Inter-epidemic virological surveillance:** Detection of circulating virus in vectors and hosts during inter-epidemic periods to clarify persistence mechanisms

9.5. Human Behavioral Dimensions

Both ABM and EBM for RVFV have largely neglected the human behavioral dimensions of pastoral management—decisions about movement timing, vaccination uptake, and veterinary service utilization. Incorporating these dimensions, potentially through agent-based behavioral models of herder decision-making [Iles et al., 2020], could improve model realism and enable evaluation of behavior-change interventions.

10. Conclusion

This review has synthesized the scientific literature on agent-based modeling and equation-based modeling of Rift Valley Fever virus transmission, with emphasis on ecosystem-integrated studies. The key conclusions are:

1. **ABM and EBM are complementary, not competing:** ABM excels at representing spatial heterogeneity, individual behaviors, and ecosystem interactions; EBM provides analytical tractability, R_0 -based thresholds, and optimal control frameworks. Integrated hybrid approaches are needed.
2. **The Ferlo ecosystem is a global model system:** Research in northern Senegal has produced the most mechanistically detailed ecosystem-integrated RVFV models, combining ABM (Paul et al.), mechanistic EBM (Durand et al.), hydrological modeling (Soti et al.), and remote sensing (Vignolles et al.) into a coherent research program.
3. **Vector parameters and rainfall are the primary drivers:** Across all modeling paradigms, mosquito biting rate, mortality, and rainfall-driven abundance emerge as the dominant determinants of epidemic potential, providing clear targets for surveillance and intervention.
4. **Ecosystem integration enhances model realism and operational relevance:** Coupling transmission models to hydrological and remote sensing data

improves predictive accuracy and enables development of operational early warning systems.

5. **Significant gaps remain:** Hybrid ABM–EBM frameworks, climate change projections, wildlife reservoir modeling, and systematic model validation against independent data are priority areas for future research.

As climate change alters the environmental conditions that drive RVFV transmission, and as livestock trade and land use change create new opportunities for geographic spread, the integration of advanced computational modeling with ecosystem data will be essential for anticipating and preventing future outbreaks.

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